Work Breakdown Structure (WBS)

Purpose

- Decompose the project into **manageable** components
- Improve **planning, scheduling, and resource allocation**

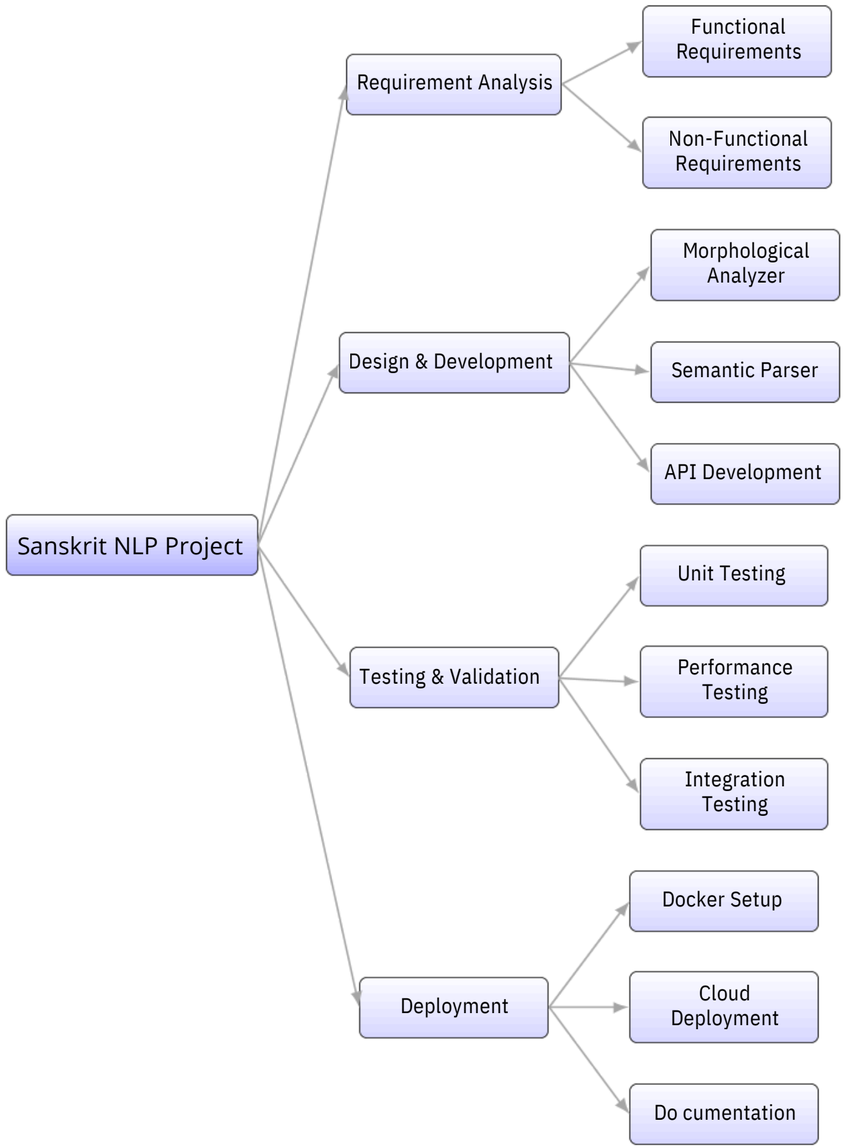
Structure

- **Level 1:** Project
- **Level 2:** Major phases **(Analysis, Development, Testing, Deployment)**
- **Level 3:** Subtasks

Management Insight

- Enables **clear task ownership**
- Supports accurate **time and cost estimation**
- Forms the basis for **PERT, CPM, and scheduling**

Work Breakdown Structure (WBS)



Critical Path Method (CPM) Analysis

Purpose

- Identify the **longest path of dependent activities**
- Determine **minimum project completion time**

Key Insight

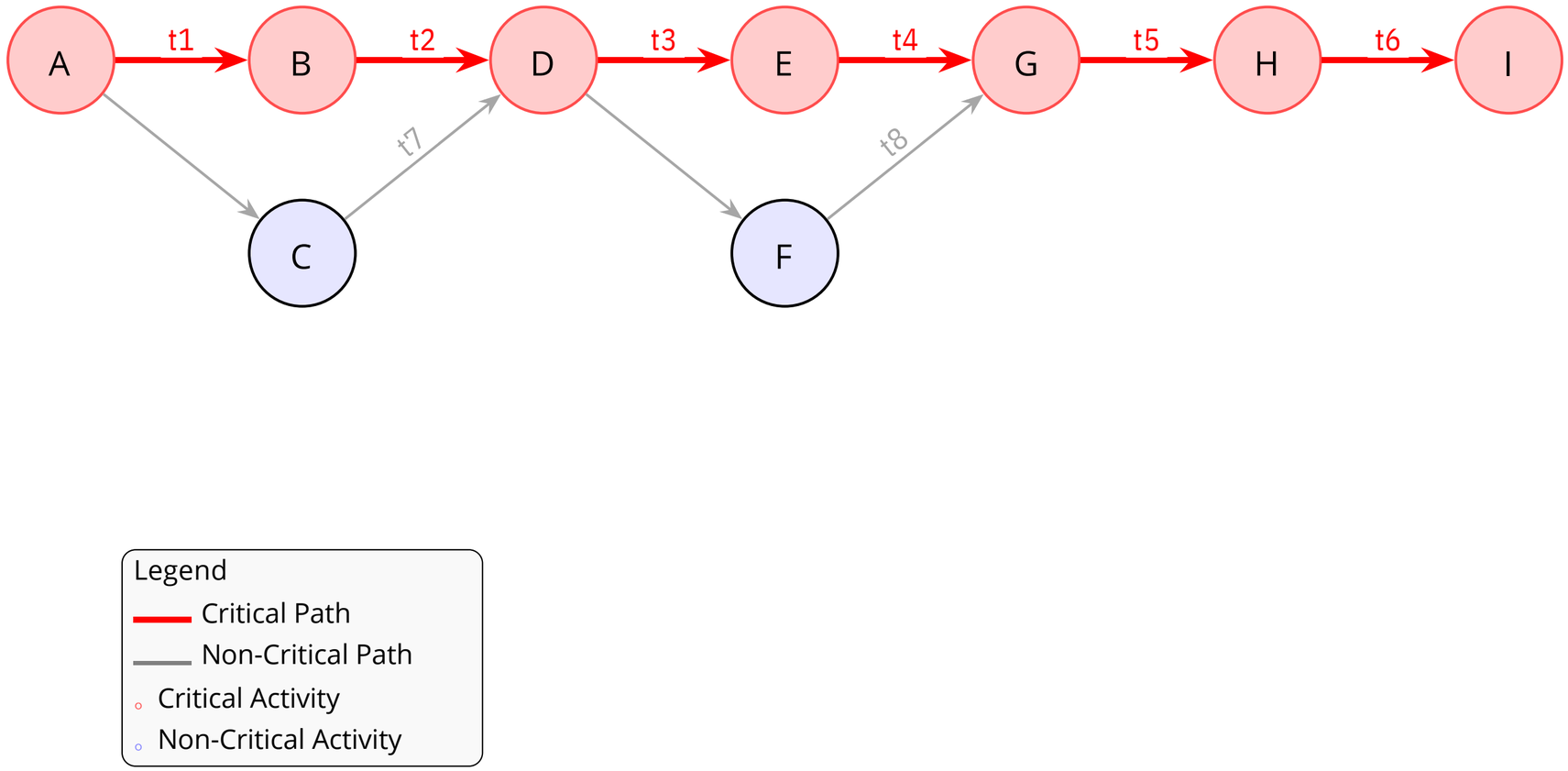
- **Critical Path: A → B → D → E → G → H → I**
- **Total Duration: 218 Days**

Management Implications

- Activities on critical path have zero slack
- Any delay directly impacts project timeline
- Enables focused monitoring and control

Critical Path Method (CPM) Analysis

Critical Path Method – Project Network



Project Schedule

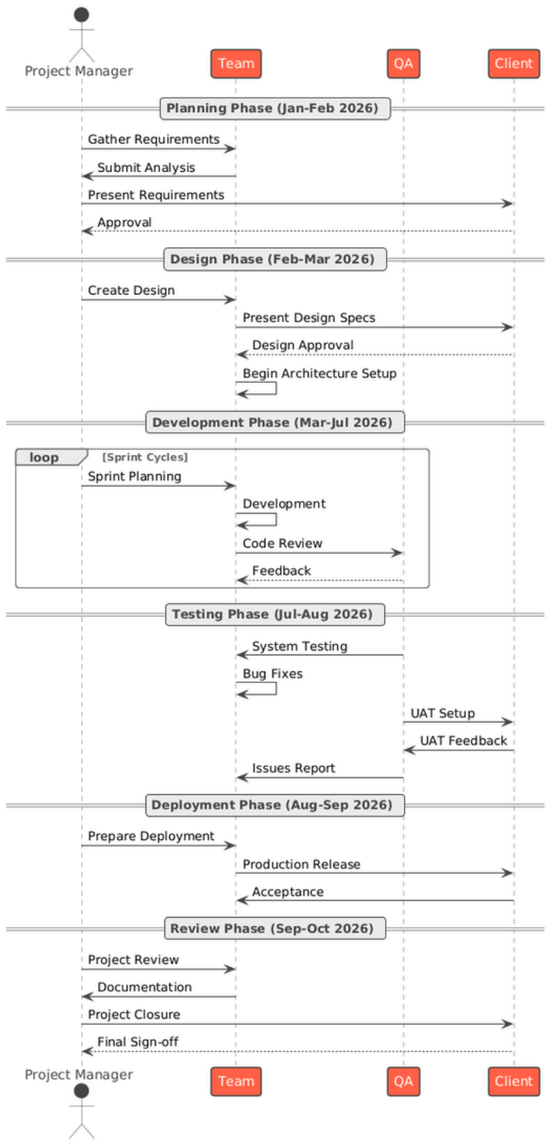
Purpose

- Visualize **project timeline and task sequencing**
- Track **start and end dates of activities**

Schedule Overview

- **Total Duration: 218 Days (~10.5 Months)**
- **Phases:**
  - **Requirement Analysis**
  - **Design & Development**
  - **Testing**
  - **Deployment**

Management Insight

- Enables progress tracking against plan
- Helps identify overlapping tasks and delays
- Supports resource planning and coordination

Project Schedule(Gantt Chart)